GlucoTwin: Multimodal Clinical Digital Twin for Type 2 Diabetes

Official 10-Slide Presentation Explanation Guide & Live Dashboard Demonstration Script

Challenge	Happiest Health · Digital Twin Challenge 2026 (Bengaluru Summit)
Team / Institution	Team Aethan · Amity University Mumbai
Team Leader	Anurag Roopchandra Kannojiya (anuragkannaujiyak@gmail.com)
GitHub Repository	https://github.com/AnuragKannojiya/Aethan_AmityUniversityMumbai
Live Workstation	Next.js 15 (port 3000) · FastAPI Microservice (port 8000)
Key Metric	Held-out ROC-AUC: 0.9925 · PR-AUC: 0.9965 · Sensitivity: 94.55% · Latency: <15ms

Part 1: Slide-by-Slide PPT Explanation Guide

This section provides an in-depth clinical, mathematical, and architectural breakdown of all 10 slides in the submission slide deck (presentation.pdf / presentation.pptx). Use these talking points during your presentation.

Slide 1: Title & Identity — *GlucoTwin: Multimodal Clinical Digital Twin for Type 2 Diabetes*

Core Focus: Team Aethan · Amity University Mumbai · Team Leader: Anurag Roopchandra Kannojiya

- Establishes GlucoTwin as a complete in-silico physiological twin rather than an isolated regression or forecasting script.
- Highlights the 2-hour ahead predictive horizon and closed-loop counterfactual simulation capability tailored for Indian Type 2 Diabetes.
- Emphasizes the multimodal fusion of static clinical history (EHR) and continuous dynamic telemetry (CGM + wearable IoT).

Slide 2: 01 — Clinical Problem (Reactive vs. Proactive) — *Reactive care sees the average. Patients live through the spikes.*

Core Focus: Why retrospective quarterly HbA1c checks fail to prevent daily postprandial vascular damage

- **The 90-Day Rear-View Mirror:** Standard diabetes care relies on quarterly HbA1c tests. An HbA1c of 7.5% merely reflects a retrospective mathematical average, completely hiding acute postprandial excursions (>180 mg/dL) and nocturnal hypoglycemia (<70 mg/dL).
- **The Asian-Indian Phenotype:** India has 101M+ diabetic adults (ICMR-INDIAB). South Asians develop severe insulin resistance and early beta-cell failure at lower BMIs (23-25 kg/m2) due to visceral adiposity, hypertriglyceridemia, and high polygenic susceptibility (TCF7L2, PPARG).
- **Vascular Toxicity:** Chronic glucose volatility causes endothelial oxidative stress, driving retinopathy, nephropathy (eGFR loss), and coronary heart disease. Standard CGMs beep only after the threshold is breached; GlucoTwin predicts it 2 hours in advance.

Slide 3: 02 — System Architecture — *Six blocks turn multimodal signals into clinical action.*

Core Focus: From Static EHR + 5-Minute Wearable IoT Streams to Real-Time Clinical Decision Support

- **Block 1 (Dual Data Ingestion):** Ingests 53 static EHR features and 54 dynamic continuous wearable channels.
- **Block 2 (Feature Pipeline & Store):** Leakage-free feature normalization, temporal rolling kinetics, and circadian feature encoding.
- **Block 3 (Hybrid Multimodal AI):** PyTorch Temporal Self-Attention LSTM (0.60) + Tabular XGBoost (0.40) ensemble.
- **Block 4 (Digital Twin Engine):** Organ strain estimation, ADA/RSSDI risk tiering, and mechanistic intervention simulation.
- **Block 5 (FastAPI Microservice):** Sub-15ms REST API powering clinical queries and in-silico what-if trials.
- **Block 6 (Next.js 15 EHR Dashboard):** 3-view closed-loop clinical platform (Physician Workstation, Patient Companion, AI Governance).

Slide 4: 03 — Data Integrity & DPDP/HIPAA Sandbox — *Two streams. One patient-level split. Zero leakage.*

Core Focus: 100% Synthetic Sandbox Compliant with India's DPDP Act 2023 & HIPAA

- **Stream 1 (Static EHR · 53 Features):** 200 synthetic Indian patients across 8 cities. Features include demographics, duration, HbA1c, fasting glucose, lipid triad (TG/HDL), eGFR, creatinine, medication classes, and polygenic markers (TCF7L2 rs7903146, PPARG rs1801282).
- **Stream 2 (Dynamic IoT · 54 Channels):** 1,728,000 continuous 5-minute telemetry rows (30 days/patient) tracking CGM glucose, heart rate, HRV RMSSD, pedometer steps, and sleep architecture.
- **Zero-Leakage Cohort Partitioning:** Strict patient-level split (140 train / 20 val / 40 test). Feature scalers are fit strictly on training patients, ensuring 100% leak-free evaluation on held-out patients.

Slide 5: 04 — Hybrid Multimodal AI — *Sequence intelligence meets clinical feature interactions.*

Core Focus: Attention-LSTM + Tabular XGBoost Ensemble (0.60 / 0.40)

- **Branch A (PyTorch Attention-LSTM · 0.60):** Ingests a 24-step (2-hour) lookback tensor (24x54) into a 2-layer Bidirectional LSTM (128 hidden units) with Temporal Self-Attention. Saliency weights highlight which past moments drive the spike.
- **Branch B (Tabular XGBoost · 0.40):** Evaluates 485 multi-scale rolling features (mean, velocity, acceleration, volatility) combined with static biomarkers. Cost-sensitive weighting penalizes missed crises.
- **Calibrated Soft Voting:** The ensemble balances deep temporal kinetics with tabular decision-boundary precision, yielding calibrated spike probabilities.

Slide 6: 05 — Empirical Validation — *Held-out validation is precise on 40 unseen patients.*

Core Focus: 57,480 unseen 2-hour evaluation windows · P161-P200 · Zero patient overlap

- **ROC-AUC of 0.9925 & PR-AUC of 0.9965:** Exceptional discriminatory capability across unseen test sequences.
- **94.55% Sensitivity (Spike Recall):** Catches over 94 out of 100 impending hyperglycemic events 2 hours ahead, preventing emergency visits.
- **96.16% Specificity:** High specificity prevents alarm fatigue, ensuring physicians and patients trust dispatched notifications.
- **Brier Score of 0.0357:** Near-perfect probability calibration between model confidence and real clinical event rates.

Slide 7: 06 — Digital Twin Engine — *Forecast the state, triage the risk, then test the intervention.*

Core Focus: Hybrid ML + Mechanistic Physiological Compartment Modeling for 'What-If' Prescriptions

- **Multi-Organ Strain Tracking:** Real-time estimation of pancreatic beta-cell load, hepatic glucose production (HGP), and autonomic tone.
- **ADA/RSSDI Clinical Stratification:** Automated classification into Low, Moderate, High, and Critical risk tiers.
- **Counterfactual 'What-If' Simulation:** Models GLUT4 non-insulin-dependent glucose uptake from a 20-min post-meal brisk walk (-48 mg/dL peak drop), pharmacotherapy timing (-38 mg/dL), and fiber pre-loads (-18 mg/dL).

Slide 8: 07 — Clinical UI · View 1 (Doctor Workstation) — *The physician moves from triage to simulation to verified action.*

Core Focus: Clean Hospital EHR Light Theme · Next.js 15 · TypeScript · Tailwind CSS · Recharts

- **Cohort Priority Queue:** Real-time triage ranking of 200 patients by 2-hour spike probability.
- **Synchronized Dual-Stream Chart:** 6h historical CGM + 2h dashed projection alongside wearable steps, HR, and HRV.
- **Multimodal Feature Attribution:** Quantifies exact Dynamic vs. Static feature impacts in mg/dL.
- **Interactive Simulator:** Sliders for walk minutes, carb load, and meds compute new trajectories in real time.

Slide 9: 08 — Clinical UI · Views 2 & 3 (Patient & Governance) — *The loop closes when the patient acts and the system verifies.*

Core Focus: From doctor simulation to patient action and cohort governance

- **View 2 (Patient Closed-Loop Companion):** Mobile-optimized view delivering physician-verified micro-nudges ('Take a 20-min walk to lower peak by 46 mg/dL'). Interactive 'Mark Walk Completed' logs feedback.
- **View 3 (Cohort & AI Model Validation):** Auditable surveillance of cohort health, live API latency (<15ms), model agreement (96.8%), and held-out confusion matrix metrics.

Slide 10: 09 — Healthcare Business Impact & Roadmap — *Proactive diabetes care can scale from sandbox to Indian hospitals.*

Core Focus: Direct Patient Benefit · Clinician Productivity · National Health Scalability

- **Endocrinologist Shortage:** India has 1 endocrinologist per 35,000 diabetic patients. GlucoTwin enables 1 clinician to proactively oversee 200+ high-risk patients, reducing triage time by over 75%.
- **Phase 2 (Hospital Pilot):** Ayushman Bharat Digital Mission (ABDM) integration via HL7 FHIR R4 for live hospital EHR ingestion.
- **Phase 3 (Wearables):** Direct webhooks for Abbott FreeStyle Libre, Dexcom G7, Apple HealthKit, and Google Health Connect.

Part 2: 12-Minute Live Dashboard Video Demonstration Script

This script is tailored specifically for a screen-recorded demonstration of the live running dashboard at **http://localhost:3000**. Follow the bracketed action cues while reading the spoken dialogue.

Part 1: Top Navigation, Global Cohort Surveillance & Triage Queue [00:00 - 01:15]

ACTION CUE: [ACTION: Open browser at http://localhost:3000 in full screen. Hover cursor over top navigation bar and triage queue.]

SPOKEN NARRATION: “Welcome to the live demonstration of GlucoTwin, our multimodal clinical digital twin for Type 2 Diabetes. Right now, you are viewing the Next.js clinical dashboard running live at localhost:3000, powered by our high-performance FastAPI backend microservice on port 8000. In the top navigation, you will notice three dedicated operational views: the Physician Workstation for clinical triage and simulation, the Patient Closed-Loop Companion for personalized behavioral nudges, and the Cohort & AI Validation dashboard for model governance. In the top-right corner, our surveillance pills summarize the active cohort: 63 Critical and 39 High Risk patients across 200 enrolled records. On the left, our Cohort Priority Triage Queue automatically sorts all patients by their 2-hour forward spike risk, allowing clinicians to filter instantly between Critical, High, and Moderate tiers or search by name, ID, or city.”

Part 2: Patient P001 — Static EHR, Labs & Polygenic Genetic Markers [01:15 - 03:00]

ACTION CUE: [ACTION: Click on patient P001 (Rajesh Sharma) in the triage queue. Point cursor to EHR Clinical Record and Polygenic Risk Card.]

SPOKEN NARRATION: “Let us select patient P001 — Rajesh Sharma, a 58-year-old male from Mumbai flagged in the Critical risk category. In the EHR Clinical Record panel on the left, we immediately see his static medical baseline: a 7-year duration of Type 2 Diabetes, and a BMI of 28.4 kg/m², which classifies him as obese under the ICMR Asian-Indian criteria. His baseline laboratory biomarkers reveal a high quarterly HbA1c of 8.7%, fasting blood glucose of 145 mg/dL, elevated triglycerides of 195 mg/dL, and a blood pressure of 138/88 mmHg. Crucially, GlucoTwin incorporates genomic risk: Rajesh carries the TCF7L2 risk variant (rs7903146) with the heterozygous CT genotype and a composite Genetic Risk Score of 0.72, which in South Asian individuals severely blunts incretin-stimulated insulin secretion. He is on Metformin and Sitagliptin with comorbid Hypertension and Dyslipidemia.”

Part 3: Synchronized Dual-Stream Telemetry & 2-Hour Forecast [03:00 - 05:00]

ACTION CUE: [ACTION: Sweep cursor across the 5 top KPI cards, then scroll to the Dual-Stream Chart.]

SPOKEN NARRATION: “Turning to the main workspace, our top KPI cards provide immediate clinical telemetry: his current CGM glucose is 162 mg/dL; our AI ensemble predicts a 2-hour peak of 218 mg/dL, crossing the 180 mg/dL threshold in approximately 35 minutes; his 2-hour spike risk is 88%, driven by a joint 86% LSTM and 91% XGBoost probability; and live wearable telemetry shows a resting heart rate of 78 bpm, an HRV of 32 ms, and zero steps for the past 45 minutes following lunch. Below, our Dual-Stream Recharts visualization displays his past 6 hours of continuous CGM readings in solid blue against the green 70 to 180 mg/dL safety band. The dashed magenta line projects the impending spike surging past 180 to 218 mg/dL. Synchronized directly beneath it, the wearable sensor panel proves that complete postprandial physical inactivity coupled with falling HRV is driving this acute excursion.”

Part 4: Interactive Counterfactual Simulation & Closed-Loop Dispatch [05:00 - 07:00]

ACTION CUE: [ACTION: Drag the 'Post-Meal Walk' slider to 20 min, check 'Administer Scheduled Dose', point to the new green trajectory, and click 'Dispatch Nudge to Patient'.]

SPOKEN NARRATION: “Now, let us experience the core innovation of GlucoTwin: the Interactive Counterfactual Simulator. Rather than passively waiting for Rajesh to suffer a spike, the clinician can test interventions in-silico. Watch what happens as I drag the Post-Meal Walk slider to 20 minutes: instantly, the Digital Twin computes the physiological response! Simulating non-insulin-mediated GLUT4 glucose uptake in skeletal muscle, a green dashed trajectory overlays the chart. His predicted peak drops from 218 down to 172 mg/dL—a 46 mg/dL peak reduction that keeps him safely inside the green euglycemic zone! When we also verify his scheduled Metformin dose, the curve flattens completely. The doctor now clicks 'Dispatch Nudge to Patient'. A verified digital prescription is dispatched in real time to Rajesh's smartphone app.”

Part 5: Multimodal Feature Attribution & LSTM Temporal Attention [07:00 - 08:30]

ACTION CUE: [ACTION: Scroll down to Multimodal Feature Attribution bars and the LSTM Temporal Attention grid.]

SPOKEN NARRATION: “Scrolling down, the Multimodal Feature Attribution panel answers the clinical question: Why does Rajesh have an 88% spike risk? The model attributes +34 mg/dL of impact to dynamic CGM velocity, +26 mg/dL to post-meal inactivity from his pedometer, +22 mg/dL to baseline HbA1c, and +14 mg/dL to his TCF7L2 genetic risk allele. On the right, the 24-Step Temporal Attention Footprint shows that the LSTM network placed over 40% of its attention weight on the rapid acceleration occurring between 35 and 50 minutes ago, providing complete clinical transparency and explainability.”

Part 6: View 2 — Patient Closed-Loop Companion & Nudge Verification [08:30 - 10:00]

ACTION CUE: [ACTION: Click '2. Patient Closed-Loop View' in the top header. Point to the green prescription card and click 'Mark Walk / Dose Completed'.]

SPOKEN NARRATION: “Let us now switch to View 2: The Patient Closed-Loop Companion. On Rajesh's mobile device, complex graphs are replaced by a reassuring, physician-verified action plan: 'Prescribed 20-min brisk post-meal walk + scheduled Metformin dose to lower 2h peak from 218 to 172 mg/dL.' Rajesh sees clearly that taking this walk blunts his peak by 46 mg/dL and prevents vascular damage. Below, his connected CGM patch and smartwatch readings are displayed. Once Rajesh completes his walk, he taps 'Mark Walk / Dose Completed'. The status immediately updates to 'Action Logged & Synced to Doctor', successfully closing the loop between doctor, digital twin, and patient!”

Part 7: View 3 — Cohort Governance & Empirical ML Benchmark Validation [10:00 - 11:30]

ACTION CUE: [ACTION: Click '3. Cohort & AI Validation' in the header. Point to the Leaderboard Table and Confusion Matrix.]

SPOKEN NARRATION: “Next, we navigate to View 3: Cohort & AI Model Validation, designed for clinical department heads and data science evaluators. Across the top, surveillance metrics monitor population health: 154 mg/dL cohort mean glucose and sub-15ms inference latency. In the Empirical Model Leaderboard, we evaluate performance across 40 completely unseen held-out patients spanning 57,480 two-hour windows: Branch A Attention-LSTM achieves an ROC-AUC of 0.9895 and sensitivity of 92.80%; Branch B XGBoost achieves an ROC-AUC of 0.9937 and accuracy of 95.63%; and the GlucoTwin Ensemble achieves a stellar ROC-AUC of 0.9925, PR-AUC of 0.9965, 94.55% sensitivity, and 96.16% specificity. The confusion matrix confirms that out of 57,480 unseen windows, 36,922 true spikes were successfully predicted 2 hours in advance with zero data leakage.”

Part 8: FastAPI Backend Architecture & Summary [11:30 - 12:00]

ACTION CUE: [ACTION: Briefly switch to http://localhost:8000/docs to show the interactive Swagger API documentation.]

SPOKEN NARRATION: “Under the hood, all predictions, cohort queries, and counterfactual simulations are served by our FastAPI microservice, documented here at /docs with full Pydantic v2 schemas and sub-15ms response times. Operating in full compliance with India's DPDP Act 2023 and HIPAA, GlucoTwin proves that proactive, digital-twin-guided diabetes care can transform clinical practice. All code, models, and docs are available at github.com/AnuragKannojiya/Aethan_AmityUniversityMumbai. Thank you!”

Part 3: Video Recording Tips & Unlisted Upload Instructions

1. **Screen Resolution:** Set your browser zoom to 90% or 100% on a 1920x1080 display so all panels render crisply without clipping.
2. **Cursor Highlighting:** In OBS Studio or Loom, enable mouse-click highlighting to guide the viewer's eyes.
3. **Audio Quality:** Use a USB condenser mic or quiet headset; maintain a confident, moderate pace (130-140 words per minute).
4. **YouTube Upload:** Set visibility to **Unlisted**, title the video *'GlucoTwin: Multimodal Digital Twin Prototype Demo — Team Aethan'*, and paste the URL into your repository README.md.